

(a) Report Structuring

Trachea and both main bronchi are open. No occlusive pathology was detected in the trachea and both main bronchi. ...

LLM

Trachea/Bronchi: -----
Heart: -----
Pleura: -----
Lungs: -----



(b) Automatic Label Extraction

Trachea/Bronchi: -----
Heart: -----
Pleura: -----
Lungs: -----
Mediastinum: -----
Esophagus: -----
Bones: -----

LLM

Question Answer Set (QAS 1)

Q1: Is there any lung nodule? **Yes**
Q2: Is there any atelectasis? **No**
Q3: Is there any pleural effusion? **Yes....**

Question Answer Set 2 (QAS 2)

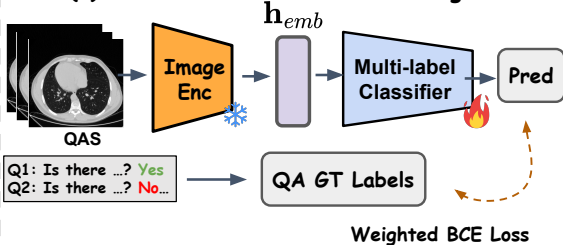
Q1: Is there pleural effusion in the right pleural space? **No**
Q2: Is there a lung nodule in the right lung? **Yes**

Question Answer Set 3 (QAS 3)

Q1: Is there ground-glass opacity in the right upper lobe (RUL)? **Yes**
Q2: Is there a lung nodule in the right middle lobe (RML)? **No**

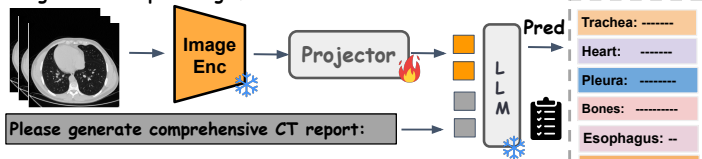


(c) Discriminative Models Training

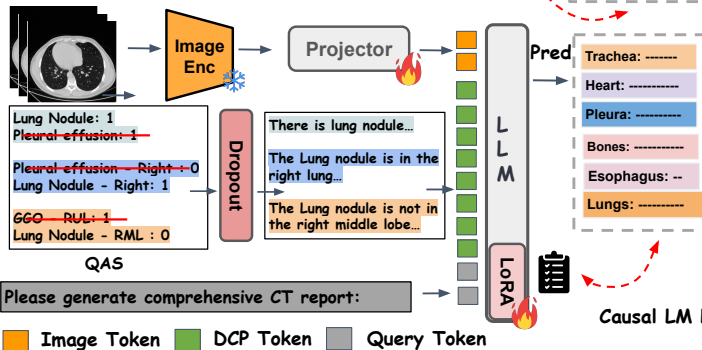


(d) Two-Stage VLM Training

Stage 1: CT-Report Alignment



Stage 2: Discriminative-guided SFT with prompt dropout



(e) VLM Inference

